

3 (a) (i) (work =) force $\times$ distance moved in the direction of the force. B1

(ii) the energy stored (in an object) due to extension / compression / change of shape B1

(b) (i) $E_K = \frac{1}{2}mv^2$ C1

$$= 0.5 \times 0.40 \times 0.30^2$$

$$= 1.8 \times 10^{-2} \text{ (J)} \quad \text{A1}$$

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(ii) (change in) kinetic energy = work done on spring / (change in) elastic potential energy C1
 $1.8 \times 10^{-2} = \frac{1}{2} \times F \times 0.080$ C1
 $F_{\text{MAX}} = 0.45 \text{ (N)}$ A1

(iii) $a = F/m = 0.45/0.40$
 $= 1.1 \text{ (ms}^{-2}\text{)}$ A1

(iv) 1. constant velocity / resultant force is zero, so in equilibrium B1

2. decelerating / resultant force is not zero, so not in equilibrium B1

(c) curved line from the origin M1
with decreasing gradient A1